Karnataka Skill Intelligence and Strategy Unit

Skillathon 2026 - Real-Data, Uncertainty-Aware Skill Gap Method

A reproducible prototype using public Karnataka data, explicit proxies, auditable linkage and Monte Carlo uncertainty.

Not an official KSI formula. All proposed indicators (including Skill Coverage Index and VOI) are analytical methods developed for this submission draft.

Reference run: 2026-10-16 **Districts:** 31 **Focus:** Tumakuru + Kalaburagi

Core principle: do not hide missing district data. Replace estimates with district-actual data when KSISU onboarding provides it; the uncertainty interval should shrink.

1. Problem and design objective

Karnataka has fragmented skilling, labour-market, industry and economic signals. The technical problem is not creating a dashboard; it is linking heterogeneous evidence at district, industry and occupation levels without pretending that incomplete data is complete.

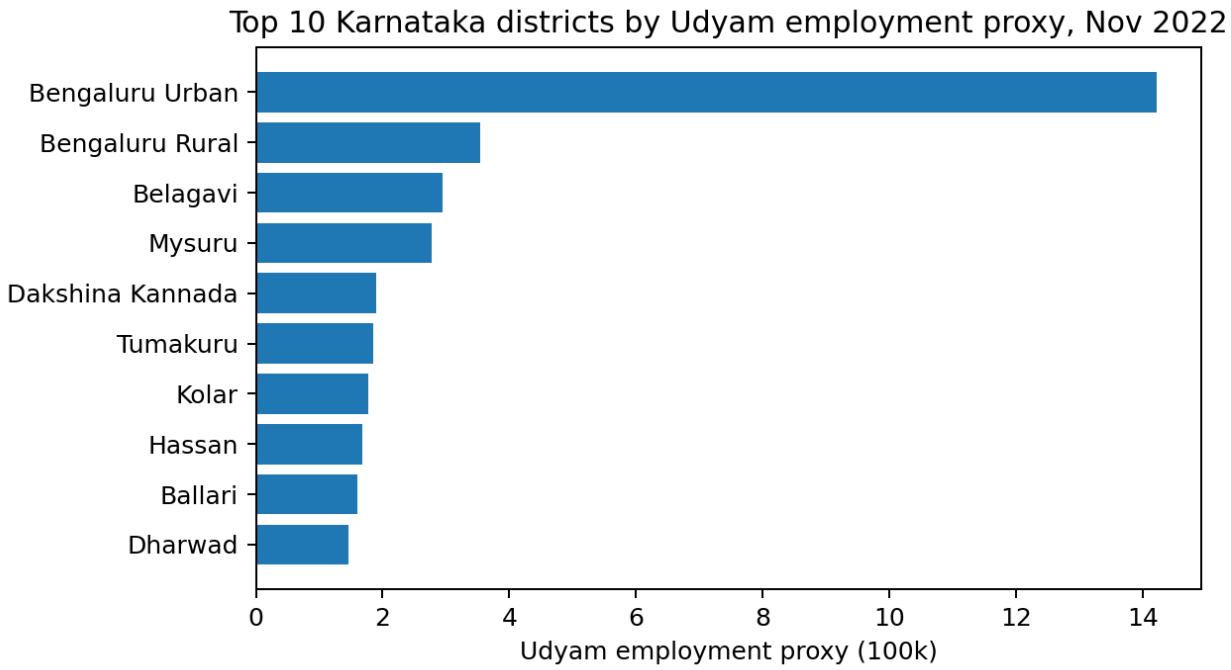
Our pipeline is: Ingest → Registry → Harmonise (LGD/NIC/NCO) → Demand model → Supply model → Alignment ratio with interval → Confidence / priority → Human review → Output.

The Skillathon brief weights Data Analysis at 70% and Tech & Integration at 30%, so the submission prioritises a credible linkage method, explicit assumptions, reproducibility and policy connection.

2. Real-data inventory and readiness

Source	Grain	Year	District match	Status
Local Government Directory - Download Directory (District, Karnataka)	district	2026	31/31 names aligned; live export not retrieved in sandbox	AMBER
Karnataka Economic Survey 2022-23 - Udyam registrations / employment up to Nov 2022	district	2022	31/31	GREEN
PLFS Annual Report 2022-23 - Karnataka manufacturing worker share	state	2022-23	N/A (state)	GREEN
DGT/Karnataka Monitoring and Evaluation Authority tracer study - ITI seats, admissions and pass-outs	division	2020-21	4 divisions only	AMBER
Karnataka Udyam district registrations - June 2026 reconciliation	district	2026	31/31	AMBER
Karnataka Economic Survey 2023-24 - Udyam employment district shares	district	2023	31/31	AMBER

Green demand evidence: Karnataka Udyam employment by district plus the state manufacturing share from PLFS. Amber supply evidence: DGT/DITE provides division-level ITI seats, admissions and pass-outs, not verified district seats.



3. Harmonisation and lineage

District matching is a cascade: (1) exact normalized name, (2) versioned alias table, (3) fuzzy match at threshold 0.90, (4) unresolved rows go to human review. Every match records method and confidence.

Industry mapping uses NIC code when available, otherwise label mapping to NIC division with review. Course/trade mapping uses a small, versioned analyst crosswalk to NCO 2-digit groups. The crosswalk is not claimed to be an official government course-to-NCO mapping.

Every raw source is registered with file_id, SHA-256 checksum, source URL, period, grain and version. Raw inputs are immutable; output rows retain source_file_ids. This creates an audit path from an indicator back to the source artifact.

4. Demand model

District employment is a proxy because reliable district PLFS estimates are not available. We use Udyam employment by district as the local industry-presence proxy and calibrate the manufacturing share with Karnataka's PLFS manufacturing worker share (10.4% in 2022-23).

For district d and occupation o: D(d,o) = E_proxy(d) × 10.4% × m(o|manufacturing). Annual openings O(d,o) = D(d,o) × (g+r), with a 5%-12% opening range. The key assumption is that the occupation mix inside manufacturing is shared across districts.

The occupation mix used in this draft is explicitly an analyst prior. It is not presented as observed PLFS microdata. The next data-improvement step is to replace it with PLFS microdata from the official microdata portal.

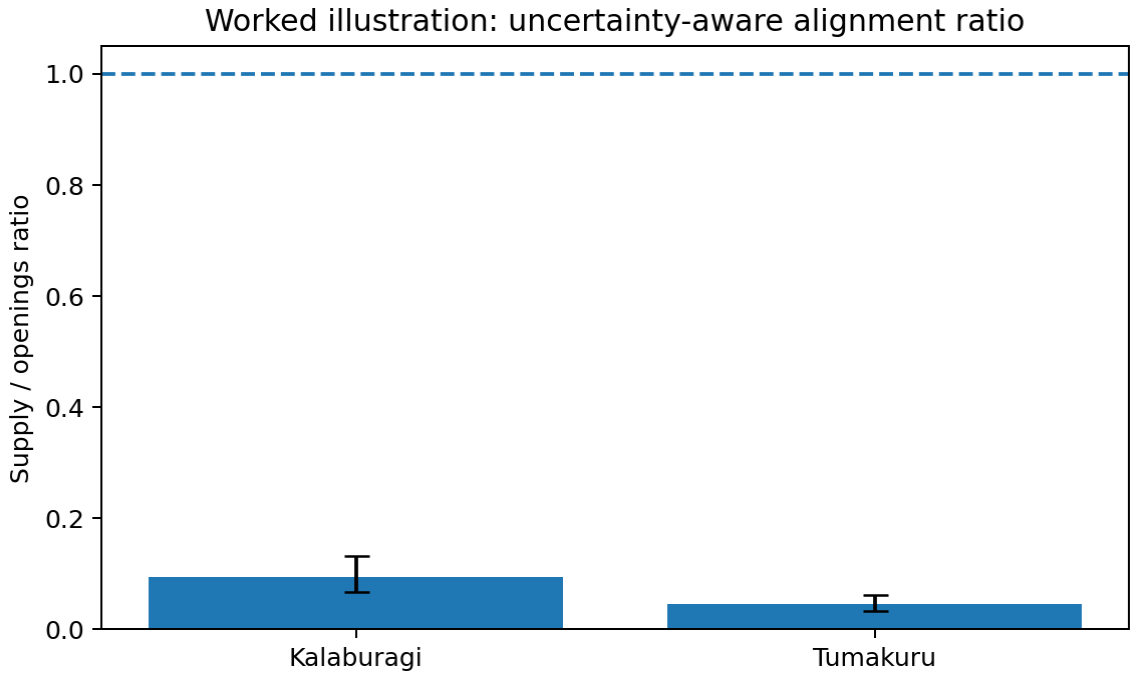
5. Supply model

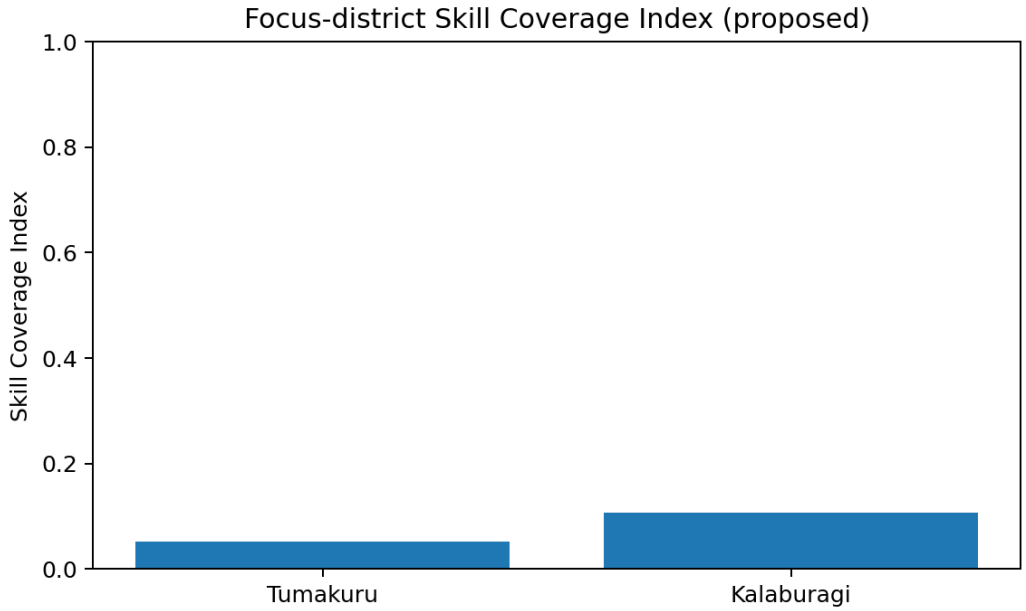
Supply keeps seats, completions and worker entry separate. Statewide ITI sanctioned seats, admissions and pass-outs from the DGT/DITE tracer study are used as a calibration layer. District seats are allocated as a transparent proxy using district Udyam employment shares because direct district seat data was not verified.

S(d,o) = seats(d) × fill rate × completion rate × sector-entry rate × x(course→NCO). The model uses triangular ranges rather than a single point estimate. The observed statewide tracer-study ratios anchor the middle of fill and completion; sector-entry remains an explicit prior because district industry placement is not directly observed.

6. Worked illustration: Tumakuru vs Kalaburagi

The two districts are shown as a contrast requested by the method design: Tumakuru as an industrial/manufacturing-linked pilot and Kalaburagi as a lower-base pilot. The numbers below are model estimates, not official district skill-gap statistics.





7. Uncertainty, confidence and data-collection priority

Each district/occupation ratio is simulated with 5,000 draws. The decision rule is conservative: if the entire 10th-90th percentile band is below 1, classify as GAP; if the entire band is above 1, classify as SURPLUS; otherwise NEEDS DATA.

The proposed Skill Coverage Index (SCI) estimates the share of local openings that local training could cover under the median ratio. The proposed Value of Information (VOI) score = interval width × annual openings, so high-stakes, high-uncertainty districts are onboarded first.

This turns data collection into a prioritised policy workflow rather than asking every district for everything at once.

8. Validation tests

Corruption test: a 31-district clean file was altered with legacy aliases, typos and 5% duplicate rows. The matcher recovered 31 of 31 canonical districts and removed 2 duplicates from 33 corrupted rows.

Sanity check: district Udyam employment shares for 2022 were compared with the published 2023 Udyam employment-share table. Median absolute difference = 0.25 percentage points. This is a temporal consistency check, not independent validation of total employment.

What still needs validation: the occupation mix, district ITI seats and industry-specific placement rate. Those are intentionally not hidden. The review queue records exactly where the evidence is weak.

9. Technology, governance and safeguards

- Public, non-personal, aggregated inputs only; no individual e-Shram/NCS records.
- Small cells can be suppressed before public release.
- Append-only audit log, source checksums and versioned crosswalks support reproducibility.
- Human review is required before a policy output is published.
- Open tooling: Python + CSV/Parquet/SQL; no vendor lock-in.
- All indicators ship with formula, source IDs, year/grain, uncertainty and limitations.

10. What KSISU gets from this approach

The output is not “a district score.” It is an evidence chain: source artifact → harmonised key → calibrated demand/supply estimate → uncertainty interval → confidence → data-collection priority → human-reviewed policy signal.

When a district provides DSDP/MIS or validated training-capacity data, the proxy is replaced rather than blended invisibly. The estimated ratio is recomputed and the interval should shrink. This makes the prototype a path to a real KSISU system instead of a one-off dashboard.

Submission note: this draft does not claim the proposed SCI or the model formula is the official Karnataka Skill Index. It is a transparent, reproducible method proposal aligned to the public challenge requirements.